1. 下列说法正确否？举例说明。

(1) $\oiint_S \vec{E}\cdot \mathrm{d}\vec{s}=0$，必有 $\vec{E}\Big|_S=0$

(2) $\vec{E}\Big|_S=0$，则 S 内必无电荷分布

(3) $\oiint_S \vec{E}\cdot \mathrm{d}\vec{s}$ 仅由 S 内包围的电荷决定

(4) $\vec{E}\Big|_S$ 仅由 S 内的电荷决定

(5) 只要 $\vec{E}$ 有对称性，就可用高斯定理求 $\vec{E}$

(6) 已知 $\vec{E}_P$，可定出 φ_P

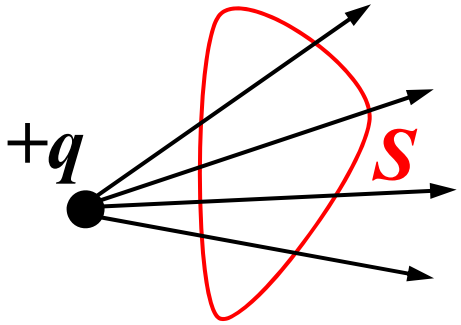
(7) 已知 φ_P，可定出 $\vec{E}_P$

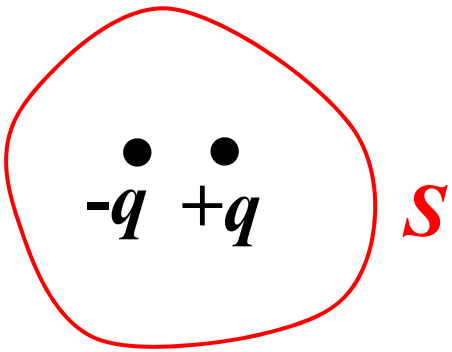
(8) $\vec{E} = \mathbf{const.} \Rightarrow \varphi = \mathbf{const.}$

(9) $\vec{E}\big|_S = \mathbf{const.}$ 不一定 $\varphi\big|_S = \mathbf{const.}$

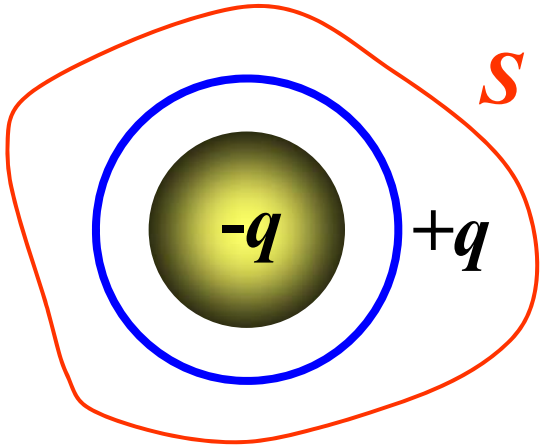
(10) $\varphi\big|_S = \mathbf{const.}$ 不一定 $\vec{E}\big|_S = \mathbf{const.}$

(1) $\oiint_S \vec{E}\cdot \mathrm{d}\vec{s}=0$，必有 $\vec{E}\Big|_S=0$ ×





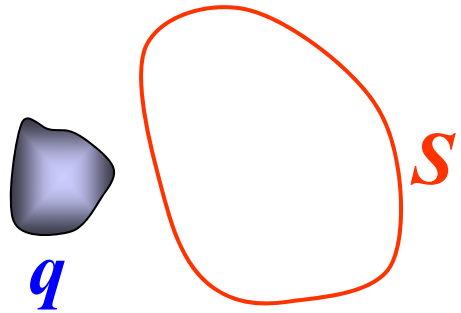
(2) $\vec{E}\Big|_S=0$, 则 S 内必无电荷分布 ×



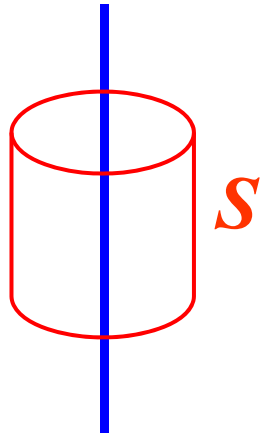
均匀带电球外有一带等量异号电荷的同心球壳。

(3) $\oiint_S \vec{E}\cdot \mathrm{d}\vec{s}$ 仅由 S 内包围的电荷决定 √

(4) $\vec{E}\big|_S$ 仅由 S 内的电荷决定 ×



S 内无电荷、S 外有电荷



S 内外均有电荷

(5) 只要$\vec{E}$有对称性，就可用高斯定理求$\vec{E}$ ×

例如有限长均匀带电直线

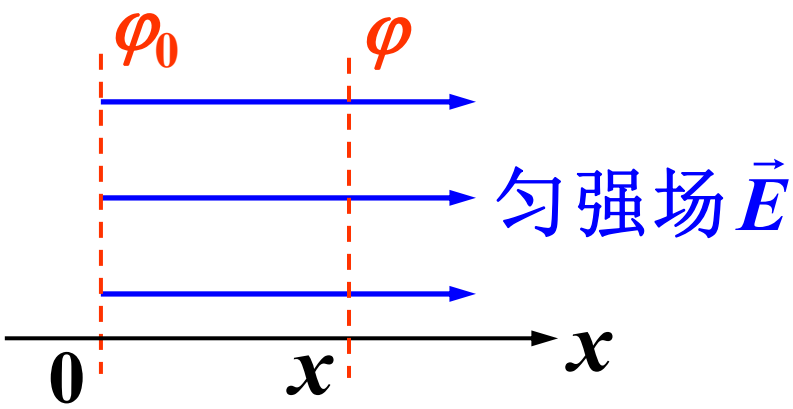
(6) 已知 $\vec{E}_P$，可定出 φ_P ×

$\varphi_P = \int_P^O \vec{E}\cdot \mathrm{d}\vec{l}$，不能用一点 $\vec{E}$ 定 φ

(7) 已知 φ_P，可定出 $\vec{E}_P$ ×

$\vec{E}_P = -\nabla\varphi\big|_P$，不能用一点 φ 定 $\vec{E}$

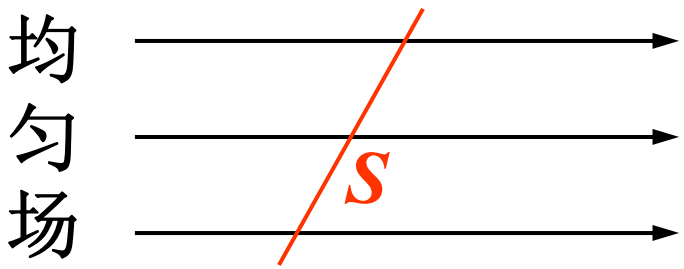
(8) $\vec{E} = \mathrm{const.} \Rightarrow \varphi = \mathrm{const.}$ ×



$$\varphi(x) = \varphi_0 - Ex$$

$$\neq \mathrm{const.}$$

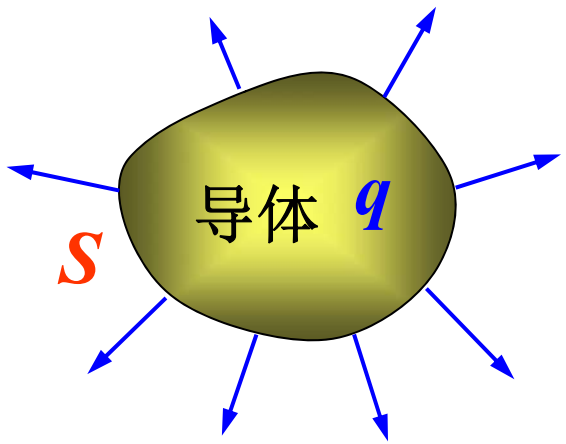
(9) $\vec{E}\big|_S = \text{const}$. 不一定 $\varphi\big|_S = \text{const}$. √



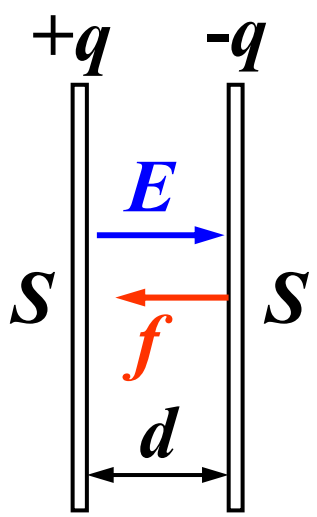
$\varphi\big|_S \neq \text{const}$.

(10) $\varphi\big|_S = \text{const}$. 不一定 $\vec{E}\big|_S = \text{const}$. √

带电导体表面 S



2. f 为两极板间的作用力大小。以下结果正确否，为何？



（$d \ll S$ 线度）

（1）根据库仑定律 $f = \dfrac{q^2}{4\pi\varepsilon_0 d^2}$ ×

（2）由电场力定义

$$f = \int_q E\mathrm{d}q = \int_q \frac{\sigma}{\varepsilon_0}\mathrm{d}q = \frac{q^2}{\varepsilon_0 S} \quad \times$$

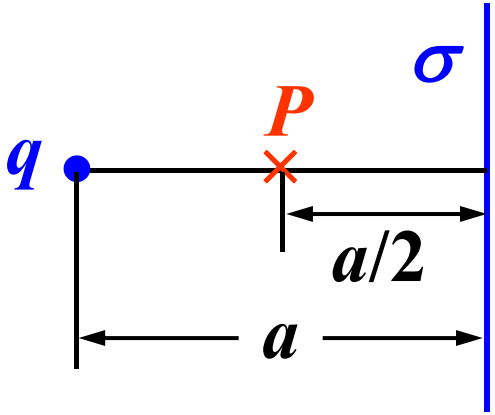
（3）由电场力定义 $f = \int_q E_+\mathrm{d}q = \int_q \dfrac{\sigma}{2\varepsilon_0}\mathrm{d}q = \dfrac{q^2}{2\varepsilon_0 S}$ √

（1）不能当点电荷计算，（2）不能用总场强计算，

（3）一板受力是另一板场强作用的结果

3. 下列做法对否？

均匀无限大带电板

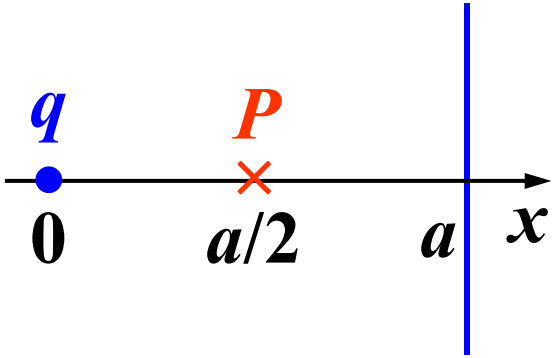


由电势叠加原理得 P 点电势：

$$\varphi_P = \frac{q}{4\pi\varepsilon_0(a/2)} - \frac{\sigma}{2\varepsilon_0}\cdot\frac{a}{2} \quad \times$$

两个电势的零点不一致。

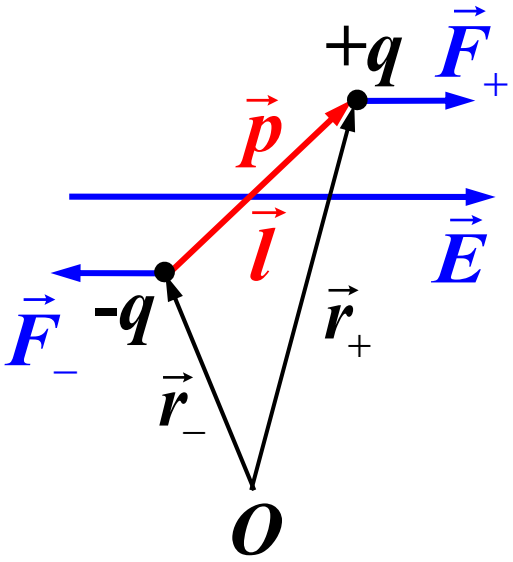
正确作法： 例如：选大平板处 $x = a$ 的电势为 0



$$\varphi_P = \int_{\frac{a}{2}}^{a} \frac{q\,\mathrm{d}x}{4\pi\varepsilon_0 x^2} + \int_{\frac{a}{2}}^{a} \left(-\frac{\sigma}{2\varepsilon_0}\right)\mathrm{d}x$$

4. 电偶极子受的力矩是对哪点而言的？

答： 电偶极子受的力矩是对任意点而言的。



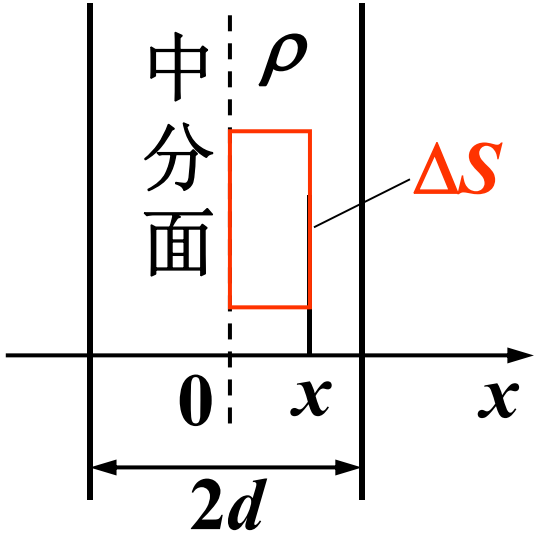
$$\vec{F}_+ = q\vec{E},\quad \vec{F}_- = -q\vec{E}$$

对任意参考点 O：

$$\begin{aligned}\vec{M} &= \vec{M}_+ + \vec{M}_- \\ &= \vec{r}_+ \times \vec{F}_+ + \vec{r}_- \times \vec{F}_- \\ &= (\vec{r}_+ - \vec{r}_-) \times q\vec{E} \\ &= \vec{p} \times \vec{E}\end{aligned}$$

结果与参考点选取无关，这是一切力偶矩的特点。

5. 求： 体电荷密度 ρ、厚 $2d$ 的无限大均匀带电平板的场强和电势分布。



解： 由对称性可知：

$$\vec{E} = E(x)\vec{i}$$

$$E(-x) = -E(x) \qquad E(0) = 0$$

对底面在中分面上的横柱面：

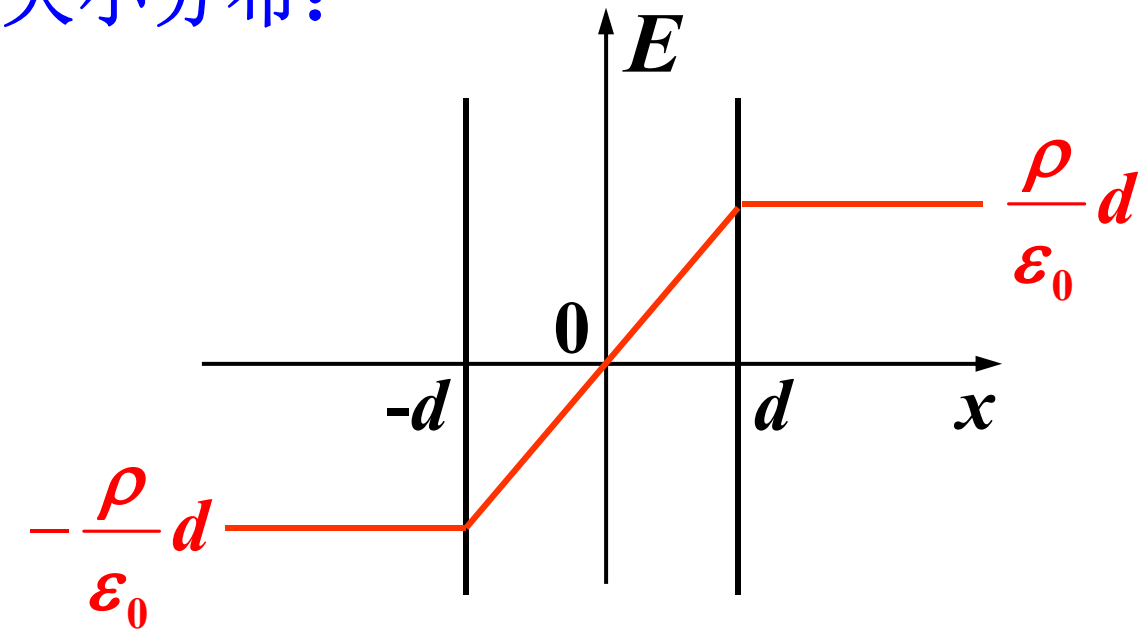
$$|x| < d: \quad \oiint_S \vec{E} \cdot \mathrm{d}\vec{s} = E(x) \cdot \Delta S = \frac{\rho(x\Delta S)}{\varepsilon_0}$$

$$\Rightarrow \vec{E} = \frac{\rho}{\varepsilon_0} x\vec{i}$$

$$|x| > d: \quad \oiint_S \vec{E} \cdot \mathrm{d}\vec{s} = E(x) \cdot \Delta S = \frac{\rho(d\Delta S)}{\varepsilon_0}$$

$$\Rightarrow \vec{E} = \frac{\rho}{\varepsilon_0} d \frac{x}{|x|} \vec{i}$$

场强大小分布：



选 $x=0$ 是电势零点

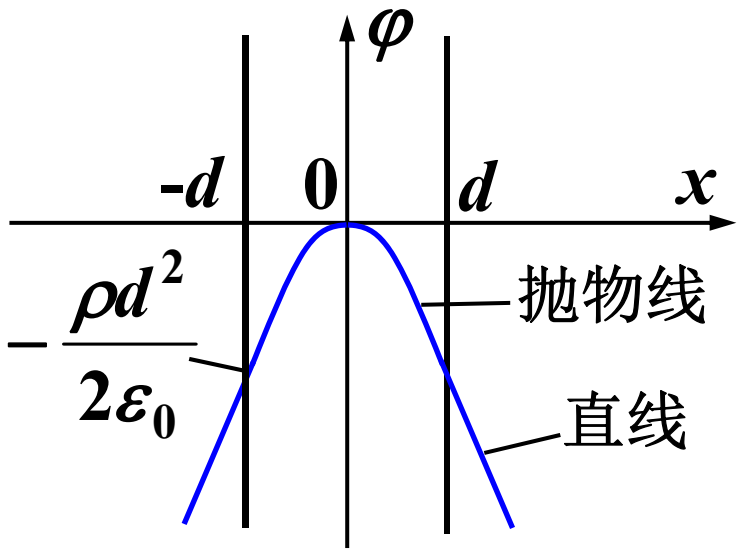
$$|x|<d:\quad \varphi=\int_x^0 \frac{\rho}{\varepsilon_0}x\,\mathrm{d}x=-\frac{\rho x^2}{2\varepsilon_0}$$

$$x>d:\quad \varphi=\int_x^d \frac{\rho d}{\varepsilon_0}\mathrm{d}x+\int_d^0 \frac{\rho x}{\varepsilon_0}\mathrm{d}x=\frac{\rho d^2}{2\varepsilon_0}-\frac{\rho d}{\varepsilon_0}x$$

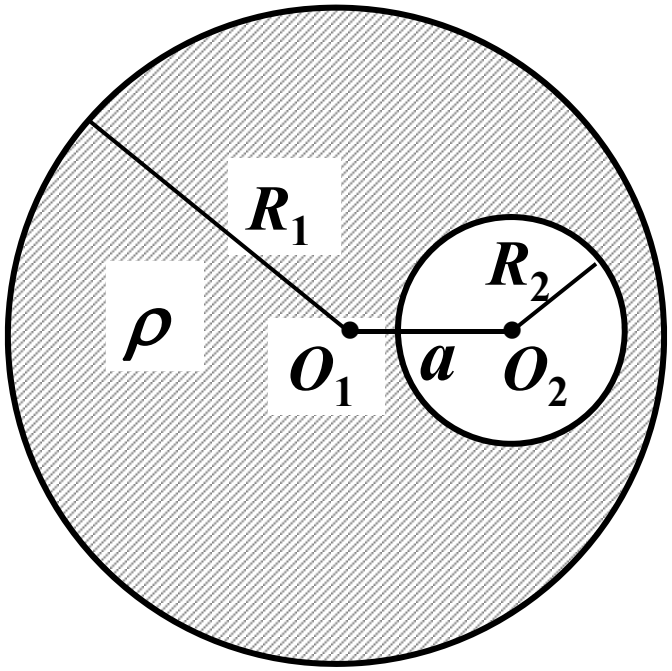
由对称性知：

$$|x|>d:$$

$$\varphi=\frac{\rho d^2}{2\varepsilon_0}-\frac{\rho d}{\varepsilon_0}|x|$$

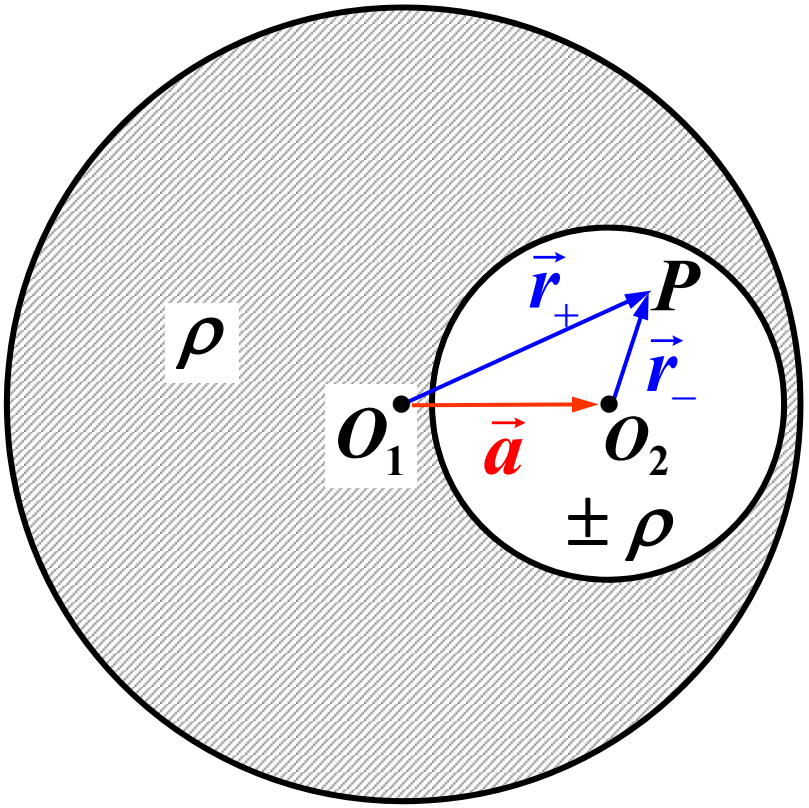


6. 在半径R_1、体电荷密度 ρ 的均匀带电球体内挖去半径 R_2 的小球体。空腔中心 O_2 与带电球心 O_1 之间的距离为 a。**求：** 空腔内任一点的场强。



解： 不能用高斯定理求解。

用叠加法： 设想在空腔处放置密度分别为 ρ 和 $-\rho$ 的均匀电荷。



$$\vec{E}_+ = \frac{\rho}{3\varepsilon_0}\vec{r}_+$$

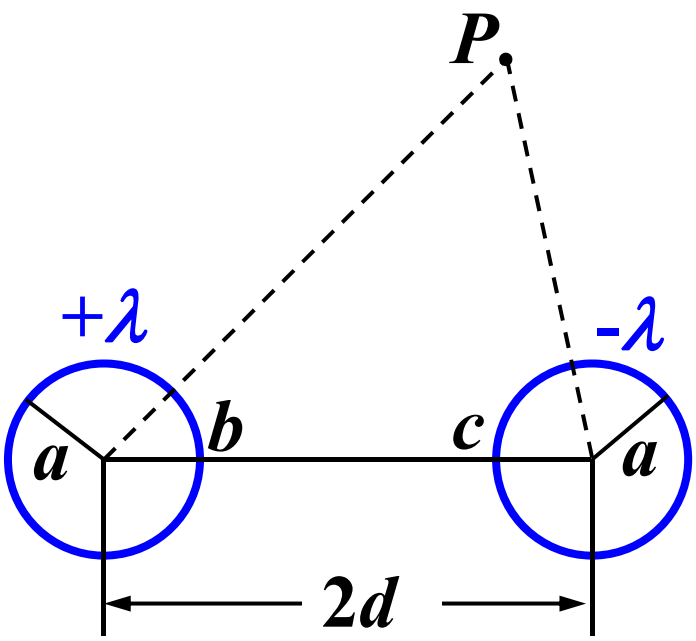
$$\vec{E}_- = \frac{-\rho}{3\varepsilon_0}\vec{r}_-$$

$$\vec{E} = \vec{E}_+ + \vec{E}_- = \frac{\rho}{3\varepsilon_0}(\vec{r}_+ - \vec{r}_-) = \frac{\rho}{3\varepsilon_0}\vec{a} = \text{const}.$$

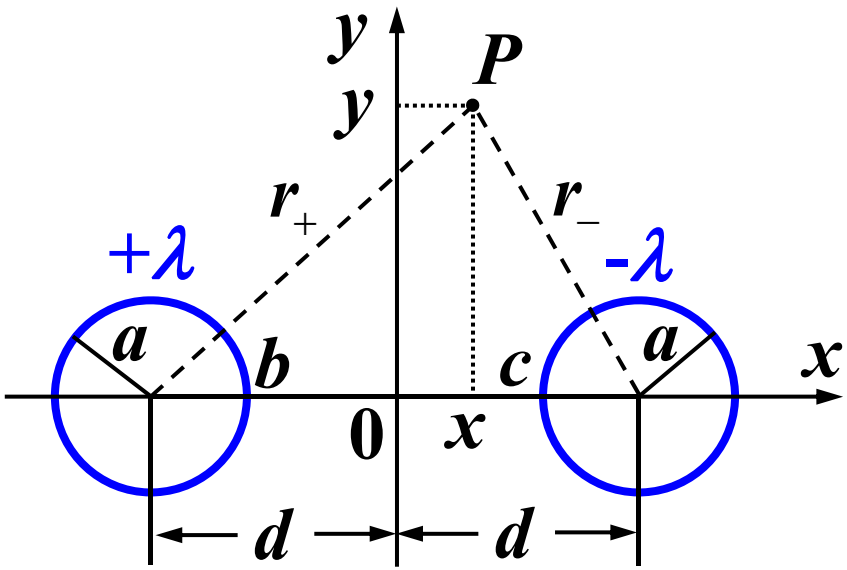
7. 两无限长均匀带电、半径 a 的平行圆柱筒，轴间距离 $2d$，电荷线密度分别为 $+\lambda$、$-\lambda$。求：

（1）圆柱筒外任一点 P 的电势 φ_P，

（2）两筒表面相距最近处的电势差 $\Delta\varphi=\varphi_b-\varphi_c$。



解：（1）求 φ_P，电势叠加法比场强积分法方便



选 0 为电势零点：

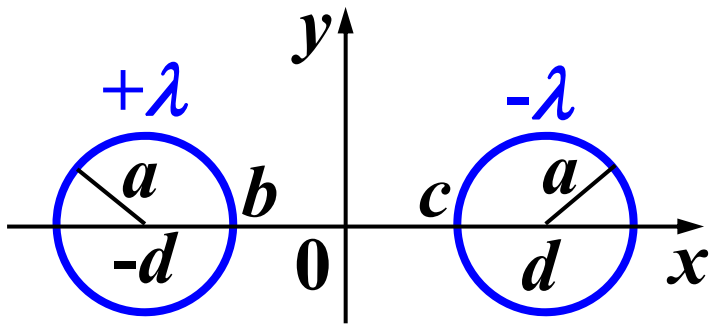
$$\varphi_+ = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{d}{r_+}$$

$$\varphi_- = \frac{-\lambda}{2\pi\varepsilon_0}\ln\frac{d}{r_-}$$

$$\varphi_P = \varphi_+ + \varphi_- = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{r_-}{r_+} = \frac{\lambda}{4\pi\varepsilon_0}\ln\frac{(d-x)^2+y^2}{(x+d)^2+y^2}$$

（2）求 $\varphi_b - \varphi_c$

$$\varphi_P = \frac{\lambda}{4\pi\varepsilon_0}\ln\frac{(d-x)^2+y^2}{(x+d)^2+y^2}$$

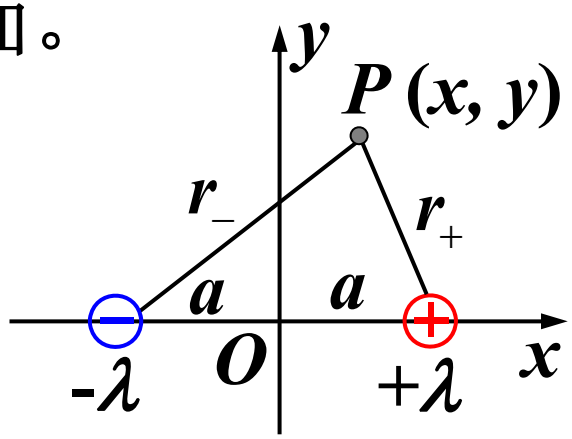


$$\varphi_b = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{2d-a}{a} \qquad \varphi_c = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{a}{2d-a}$$

$$\varphi_b - \varphi_c = \frac{\lambda}{\pi\varepsilon_0}\ln\frac{2d-a}{a}$$

【思考】用场强积分法求 $\varphi_b - \varphi_c$

8. 两平行的无限长均匀带电直线相距 $2a$，线电荷密度 $\pm\lambda$，证明等势面为圆柱面。



证明：选原点 O 为电势零点：

$$\varphi_{\pm}=\frac{\pm\lambda}{2\pi\varepsilon_0}\ln\frac{a}{r_{\pm}}$$

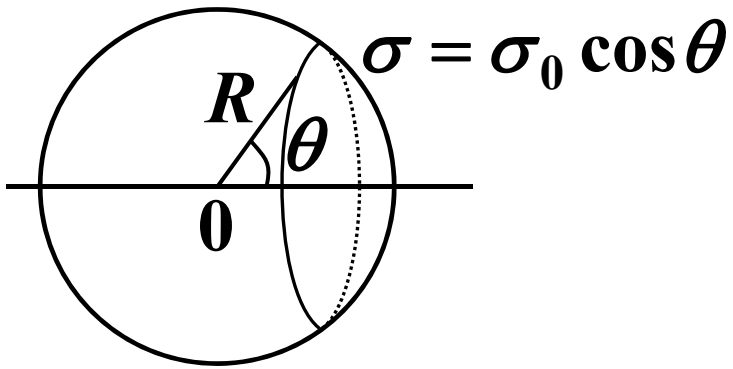
$$\varphi_P=\varphi_{+}+\varphi_{-}=\frac{\lambda}{2\pi\varepsilon_0}\ln\frac{r_{-}}{r_{+}}$$

等势面条件给出：$\dfrac{r_{-}}{r_{+}}=C=\dfrac{\sqrt{(x+a)^2+y^2}}{\sqrt{(x-a)^2+y^2}}$

$$\Rightarrow\ (x-\frac{1+C^2}{1-C^2}a)^2+y^2=(\frac{2Ca}{1-C^2})^2$$

9. 球面的面电荷分布如图，

求： 电场和电势分布



解： 叠加法

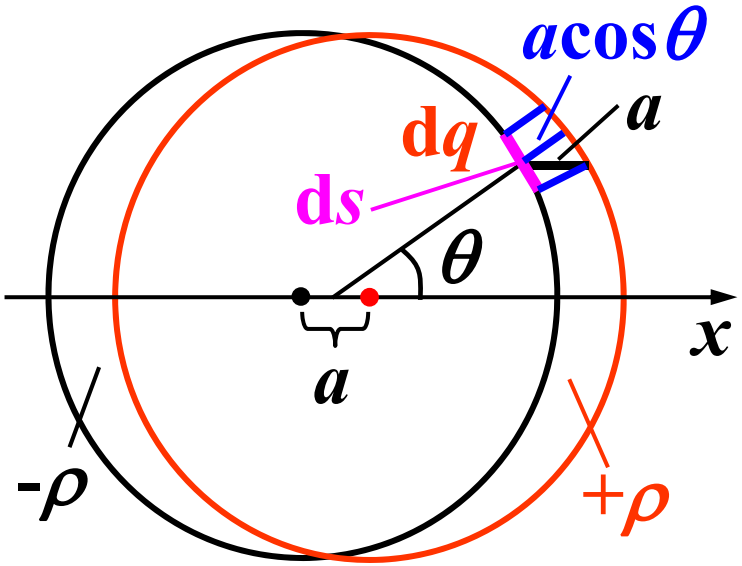
将原面电荷分布用两个半径 R、电荷密度 $\pm\rho$、错开微小距离 a 的均匀带电球的电荷分布等效。

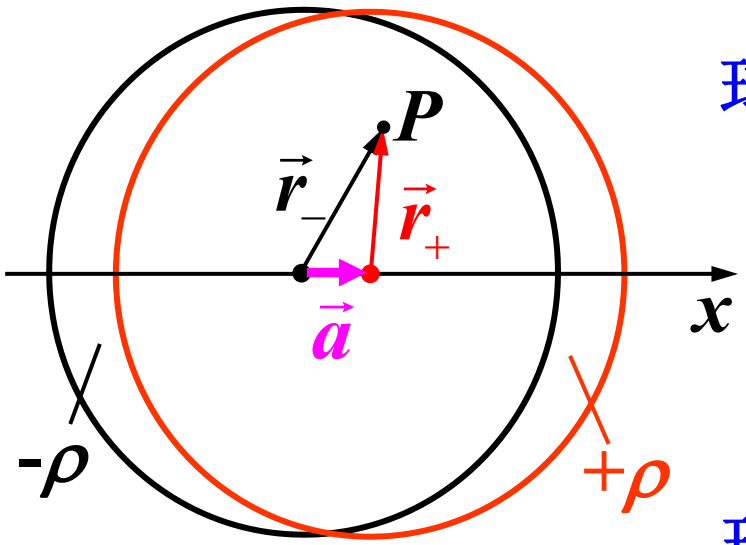
$$\mathrm{d}q = \rho \cdot (a\cos\theta \cdot \mathrm{d}s)$$

$$\sigma = \mathrm{d}q/\mathrm{d}s = \rho \cdot a\cos\theta$$

$$= \sigma_0 \cos\theta$$

$$\Rightarrow \sigma_0 = \rho \cdot a$$

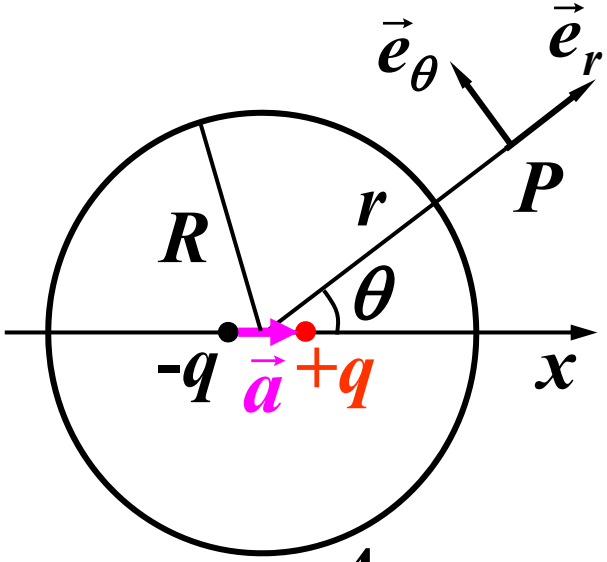




球内： $\vec{E}_{内} = \dfrac{\rho}{3\varepsilon_0}\vec{r}_+ + \dfrac{-\rho}{3\varepsilon_0}\vec{r}_-$

$$= -\frac{\rho}{3\varepsilon_0}\vec{a} = -\frac{\sigma_0}{3\varepsilon_0}\vec{i}$$

球外： 偶极子场（$a << R$）



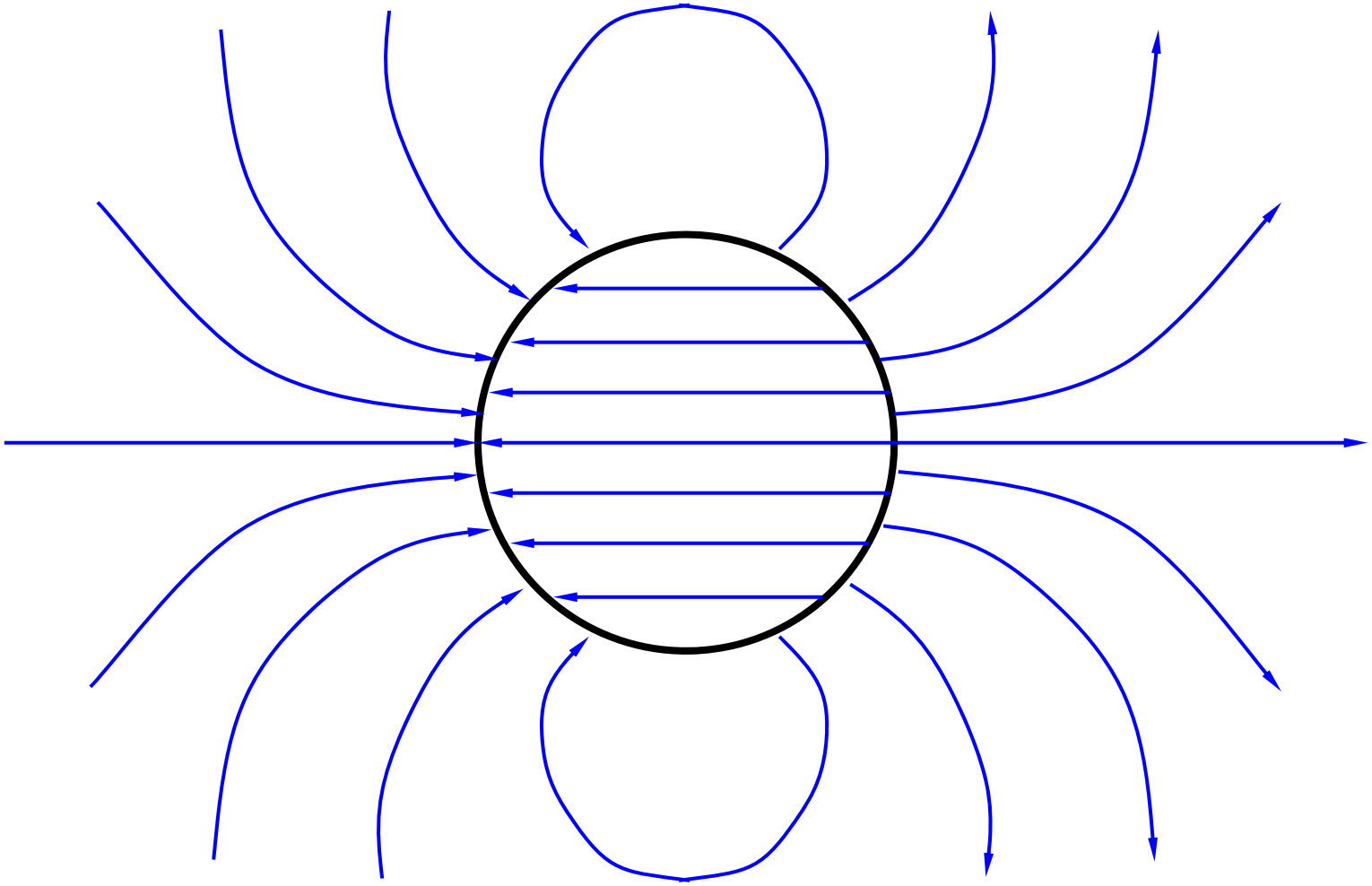
$$q = \frac{4\pi}{3}R^3\rho$$

$$\vec{p} = q\vec{a} = \frac{4\pi\sigma_0}{3}R^3\vec{i}$$

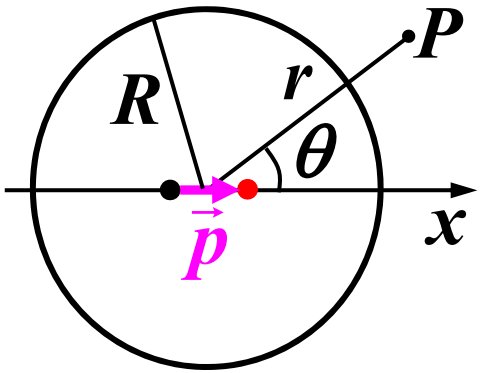
$$\vec{E}_{外} = \frac{1}{4\pi\varepsilon_0 r^3}(3\vec{p}_r - \vec{p})$$

$$= \frac{\sigma_0 R^3}{3\varepsilon_0 r^3}(2\cos\theta\cdot\vec{e}_r + \sin\theta\cdot\vec{e}_\theta)$$

电场线的分布：



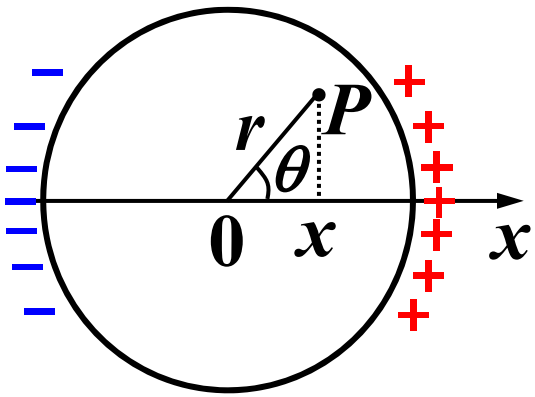
φ 的分布：



球外：偶极子 $\vec{p}=\dfrac{4\pi\sigma_0}{3}R^3\vec{i}$ 电势

$$\varphi_{外}=\frac{p\cos\theta}{4\pi\varepsilon_0 r^2}=\frac{\sigma_0 R^3\cos\theta}{3\varepsilon_0 r^2},\ \varphi_\infty=0$$

球内：匀强场 $\vec{E}_{内}=-\dfrac{\sigma_0}{3\varepsilon_0}\vec{i}$ 电势



$$\varphi\big|_{x=0}=\varphi_\infty=0$$

$$\varphi_{内}=\frac{\sigma_0}{3\varepsilon_0}x=\frac{\sigma_0}{3\varepsilon_0}r\cos\theta$$

1. 不带电小导体球 A 靠近带正电大导体球 B 时，以下结果对错，理由？

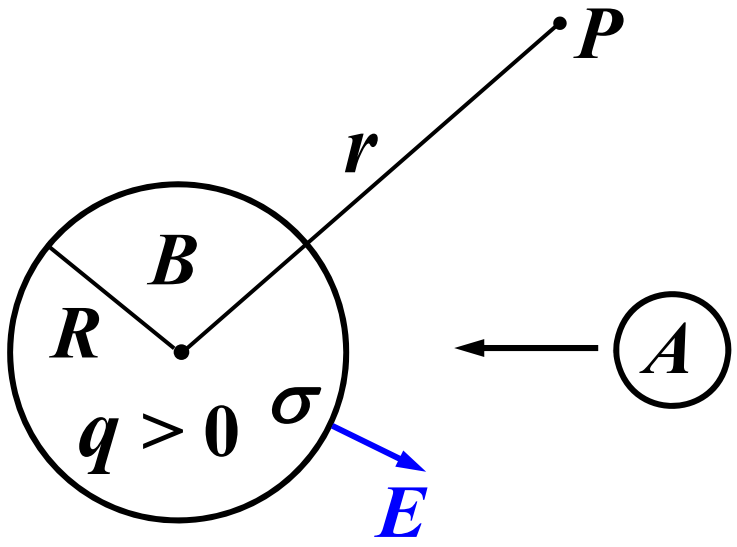
（1）$\varphi_B > \varphi_A$

（2）$\varphi_A < 0$

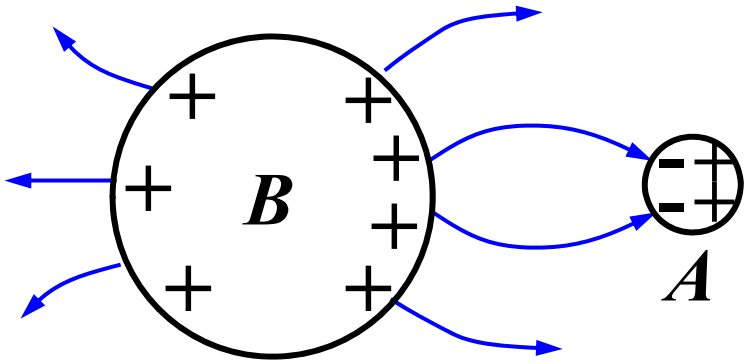
（3）$r >> R$ 时，

$$E_P \approx \frac{q}{4\pi\varepsilon_0 r^2}$$

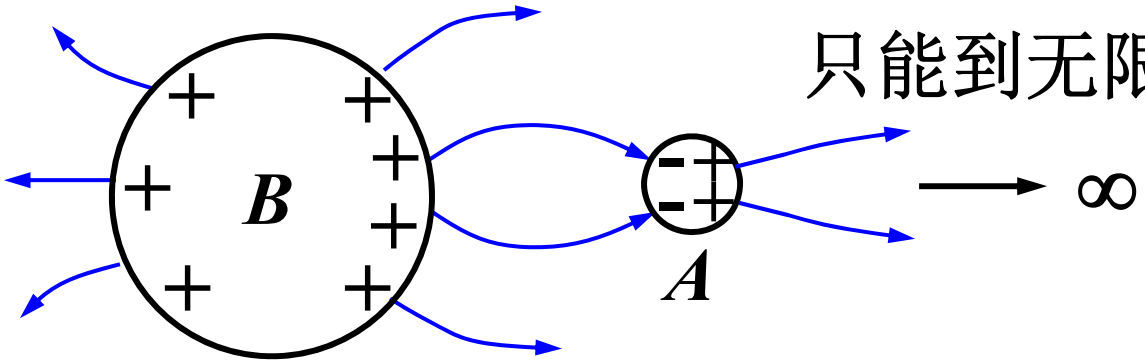
（4）$E = \dfrac{\sigma}{\varepsilon_0}$，$\sigma = \dfrac{q}{4\pi\varepsilon_0 R^2}$



（1）$\varphi_B > \varphi_A$ √ 由电场线可判断：



（2）$\varphi_A < 0$ ×



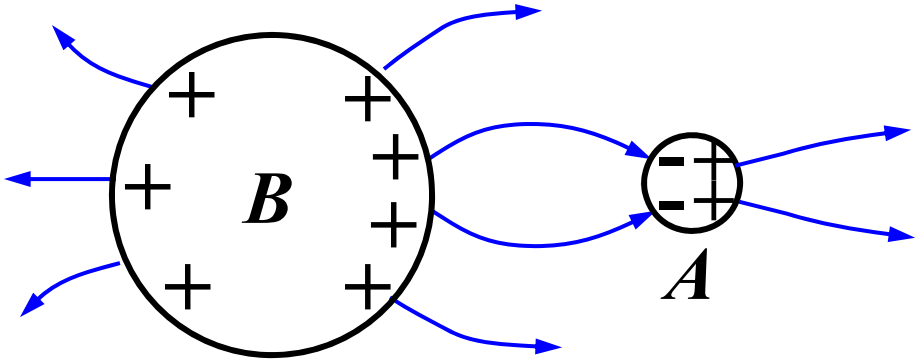
只能到无限远：$\varphi_A > \varphi_\infty$

$\longrightarrow \infty$

（3）$r >> R$ 时， $E_P \approx \dfrac{q}{4\pi\varepsilon_0 r^2}$ √

此时 A、B 大小可忽略 — 点电荷

【思考】再精确些应如何？

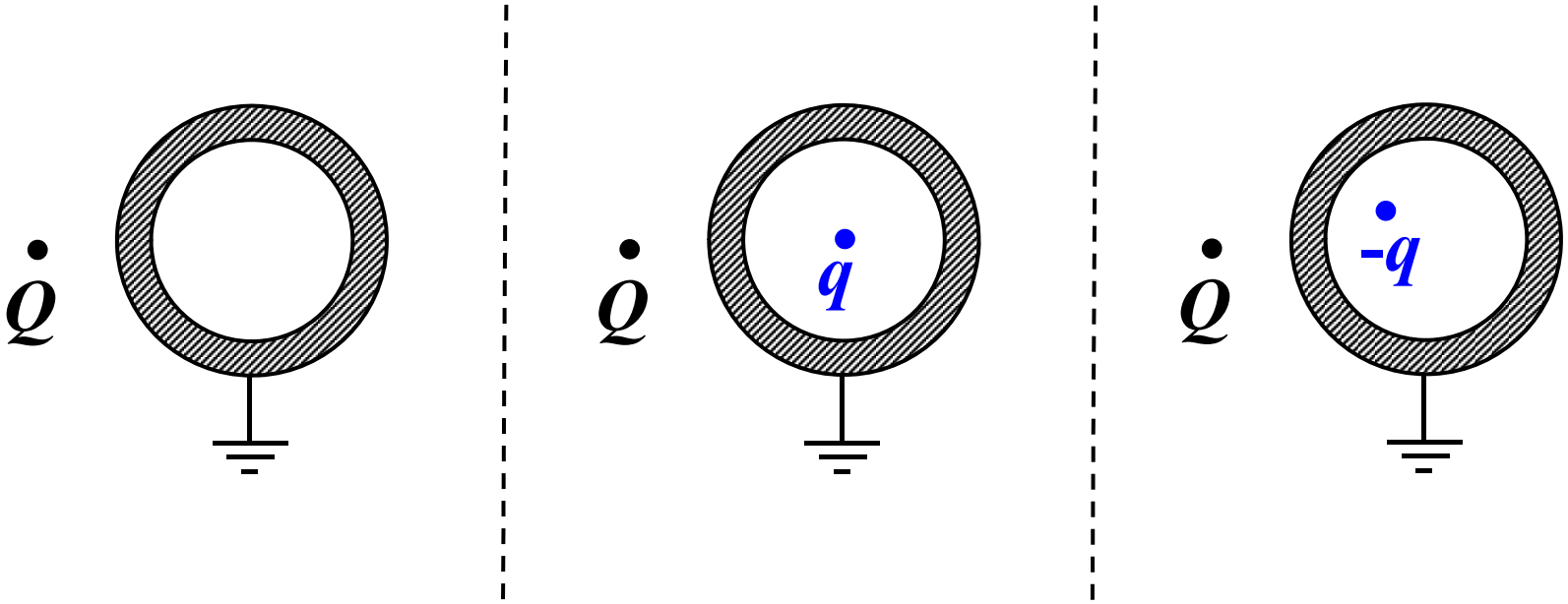


可视为点电荷与偶极子的电场的叠加。

（4） $E = \dfrac{\sigma}{\varepsilon_0}$， √ $\sigma = \dfrac{q}{4\pi\varepsilon_0 R^2}$ ×

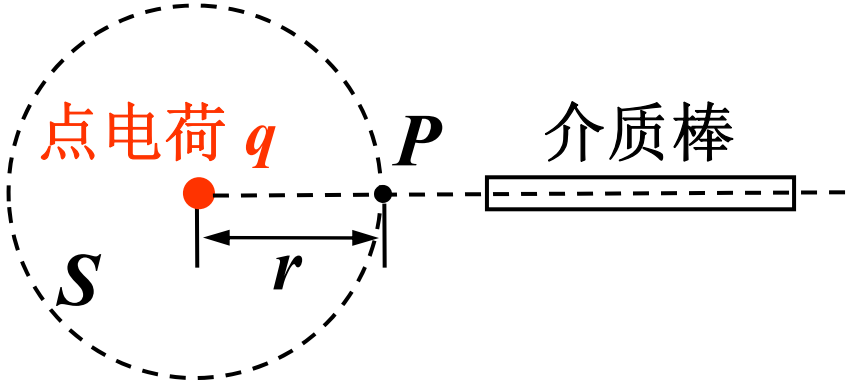
B 表面电荷分布不均匀。

2. 三种情况下电荷 Q 的电量、位置相对导体壳一样，则三者在导体壳外的电场如何？



答： 三种情况下导体壳外区域的定解条件相同，由唯一性定理，壳外电场分布也相同。

3. 下列方法求 P 点场强对否：



选高斯面：球面 S

$$\oiint_S \vec{D}\cdot \mathrm{d}\vec{s} = q$$

$$\Rightarrow D\cdot 4\pi r^2 = q$$

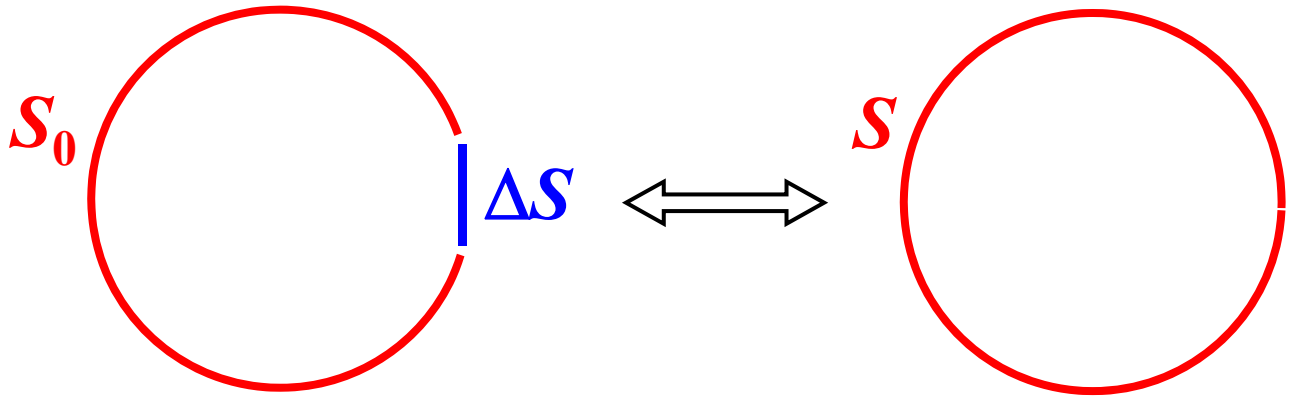
$$\Rightarrow D = \frac{q}{4\pi r^2} \quad \Rightarrow E = \frac{D}{\varepsilon_0} = \frac{q}{4\pi\varepsilon_0 r^2}$$

答：不对。介质棒表面、体内可能有极化电荷，系统不具有球对称性，S 面上各点 D 不同，不能直接用高斯定理求出 D。

4. 求均匀带电球面单位面积受力。

解： 方法一：场强叠加法

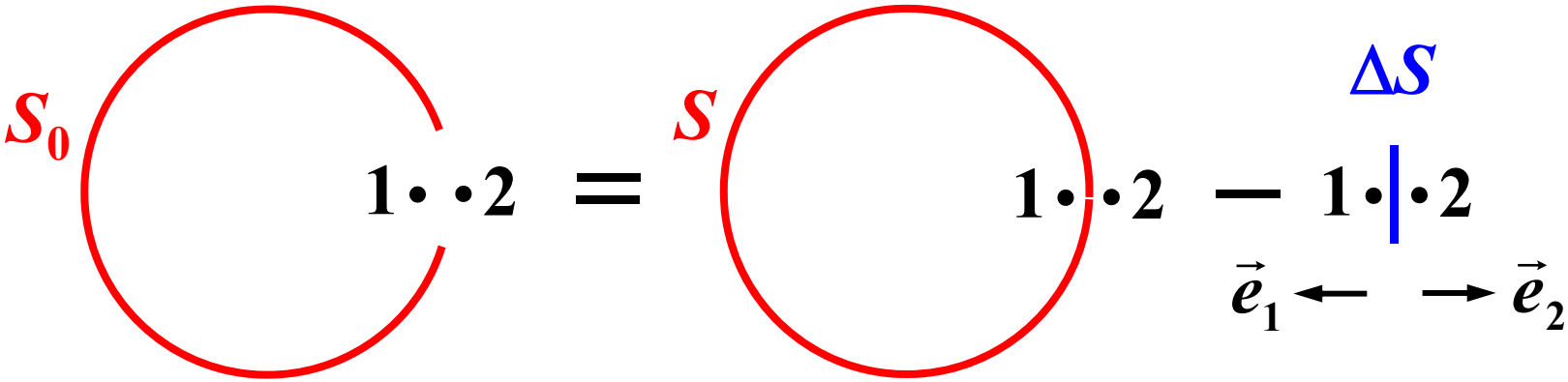
思路：不对称带电体用对称带电体的组合等效



关键：求 S_0 在单位面积 ΔS 处的场强

要点：S_0 在单位面积 ΔS 两侧的场强相等，等于 S_0 在单位面积 ΔS 处的场强。为何？

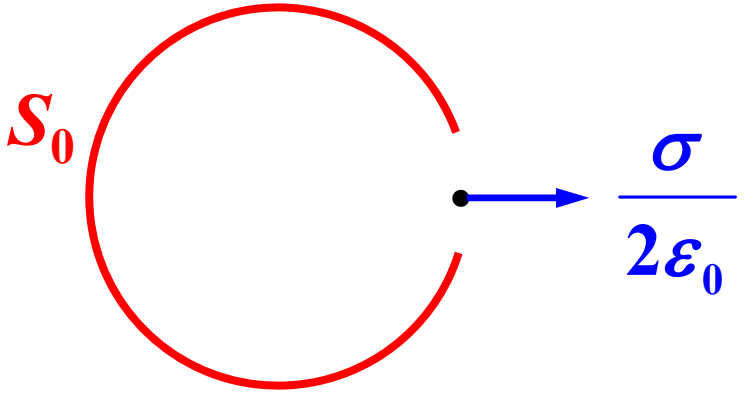
ΔS 两侧场强关系



技巧：1、2 点到 ΔS 距离 << ΔS 线度 << S_0 线度

这样 ΔS 可当作是无限大均匀带电平面。

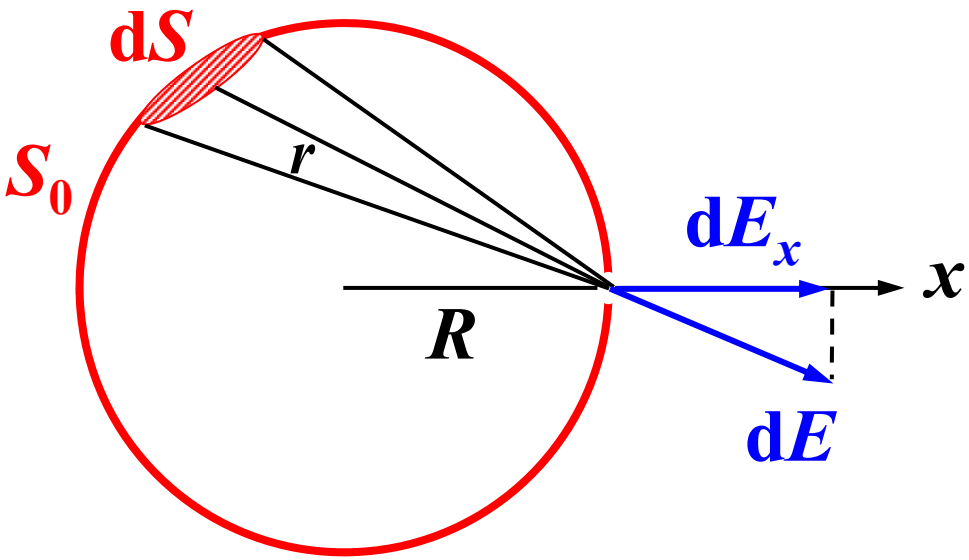
$$\left.\begin{aligned}\vec{E}_{S_0}^{1} &= \vec{E}_{S}^{1} - \vec{E}_{\Delta S}^{1} = 0 - \frac{\sigma}{2\varepsilon_0}\vec{e}_1 = -\frac{\sigma}{2\varepsilon_0}\vec{e}_1 \\ \vec{E}_{S_0}^{2} &= \vec{E}_{S}^{2} - \vec{E}_{\Delta S}^{2} = \frac{\sigma}{\varepsilon_0}\vec{e}_2 - \frac{\sigma}{2\varepsilon_0}\vec{e}_2 = \frac{\sigma}{2\varepsilon_0}\vec{e}_2\end{aligned}\right\}\text{相等}$$



S_0 在 ΔS 处的场强：$\dfrac{\sigma}{2\varepsilon_0}$

单位面积受力：$f = \dfrac{\sigma^2}{2\varepsilon_0}$

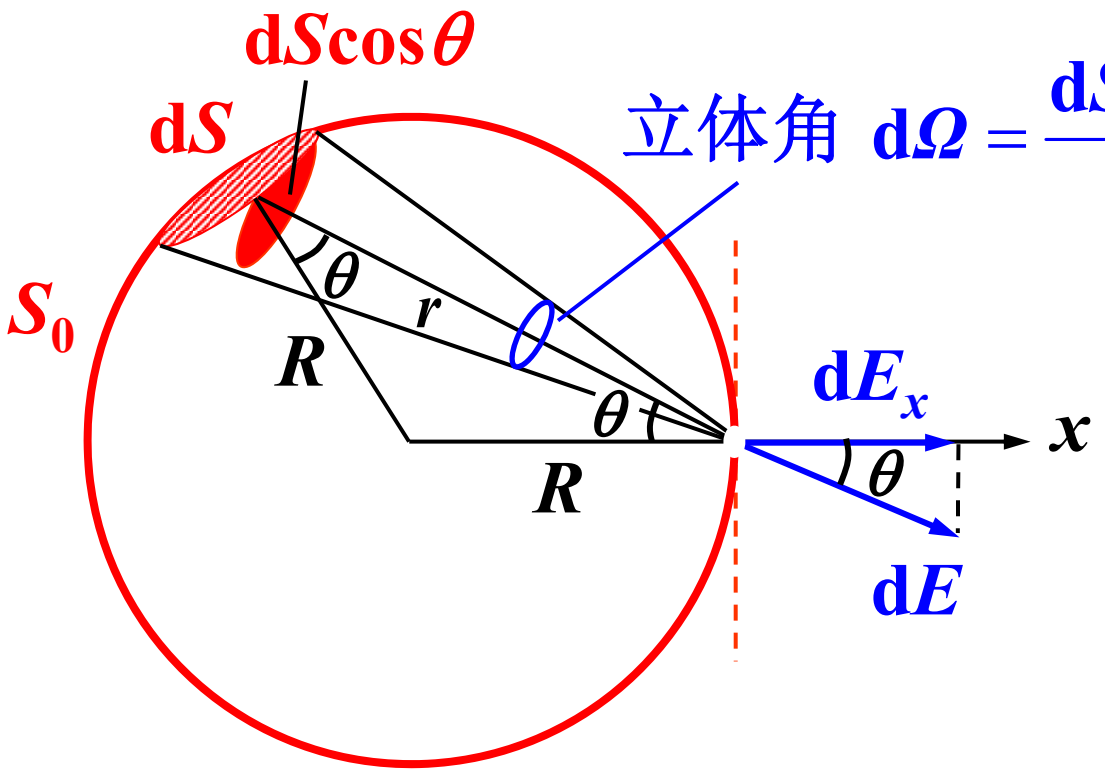
方法二：电荷元积分



$$\mathrm{d}E = \frac{\sigma \mathrm{d}S}{4\pi\varepsilon_0 r^2}$$

轴对称：

$$E = \int \mathrm{d}E_x$$



立体角 $\mathrm{d}\Omega = \dfrac{\mathrm{d}S\cos\theta}{r^2}$ 与 r 无关

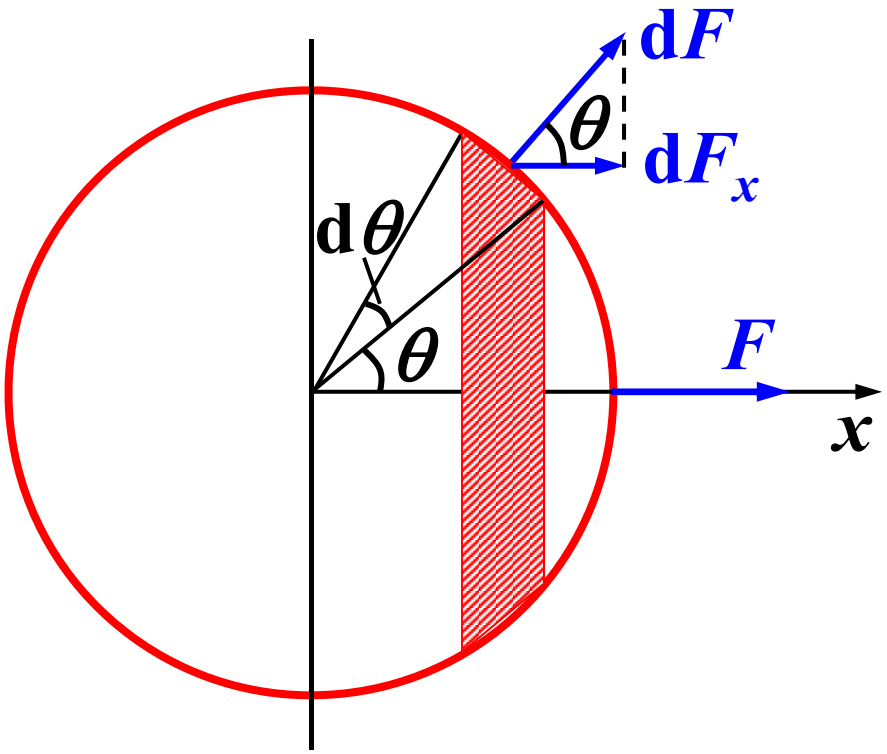
$$\mathrm{d}E_x = \mathrm{d}E\cos\theta = \frac{\sigma\mathrm{d}S}{4\pi\varepsilon_0 r^2}\cos\theta = \frac{\sigma}{4\pi\varepsilon_0}\mathrm{d}\Omega$$

$$E = \int \mathrm{d}E_x = \int_0^{2\pi}\frac{\sigma}{4\pi\varepsilon_0}\mathrm{d}\Omega = \frac{\sigma}{2\varepsilon_0} \quad \Rightarrow \quad f = \frac{\sigma^2}{2\varepsilon_0}$$

5. 求半径 R、电量 q 的导体球两半之间的作用力

解： 导体球表面单位面积受力 $f=\dfrac{\sigma^2}{2\varepsilon_0}$

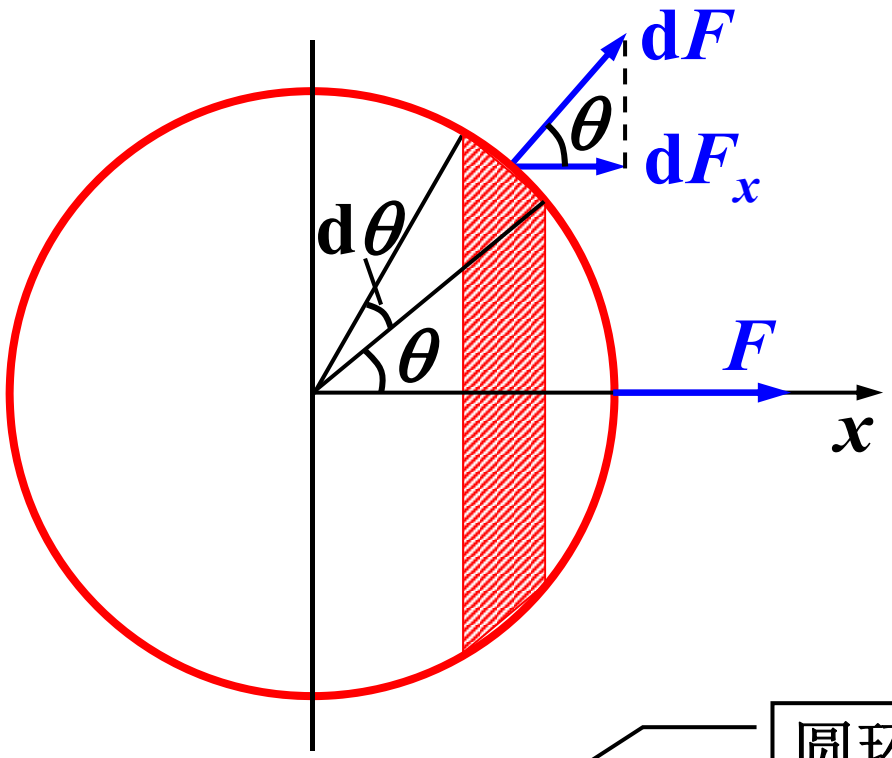
由对称性选右半球阴影部分的圆环面作分析：



$\mathrm{d}F$：球面其余部分对圆环面作用力

F：左半球面对右半球面作用力

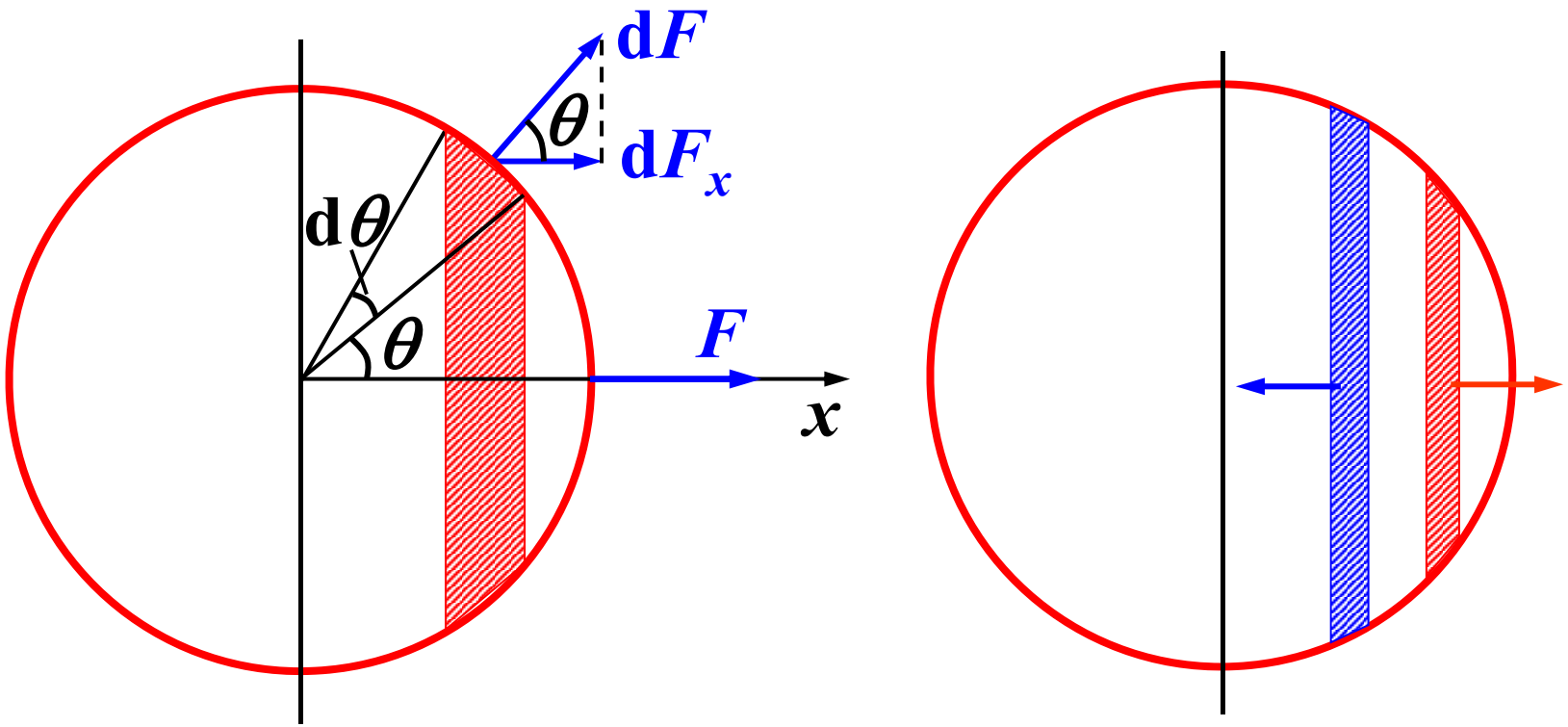
$$F=\int_{\text{右半球面}}\mathrm{d}F_x$$



圆环面积

$$\mathrm{d}F_x = \frac{\sigma^2}{2\varepsilon_0} \cdot (2\pi R \sin\theta \cdot R\,\mathrm{d}\theta) \cdot \cos\theta$$

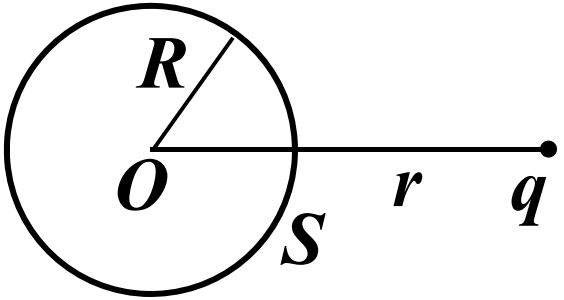
$$F = \int \mathrm{d}F_x = \frac{\pi R^2 \sigma^2}{\varepsilon_0} \int_0^{\pi/2} \sin\theta \cos\theta \mathrm{d}\theta = \frac{q^2}{32\pi\varepsilon_0 R^2}$$



【思考】用 $F = \int_{\text{右半球面}} \mathrm{d}F_x$ 计算的力是否包括右半球其它部分对右半球的作用力？

不包括。右半球电荷间静电力成对，积分抵消了。

6. 求： 导体球电势 φ



不带电导体球

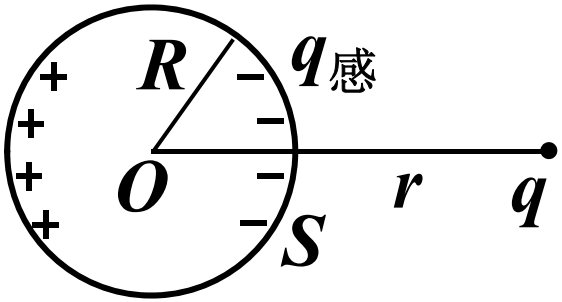
解： **方法一：叠加法**

导体球是等电势体：

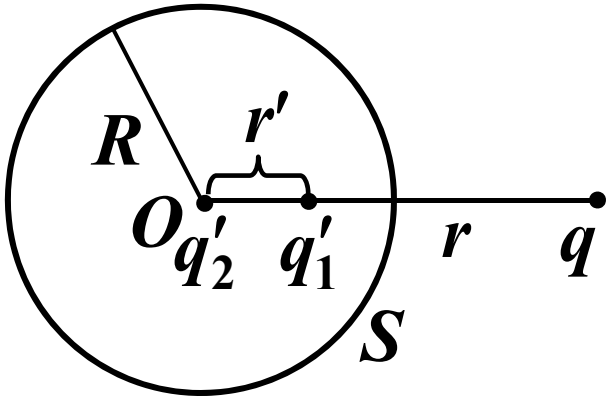
$$\varphi = \varphi_O$$

$$= \frac{q}{4\pi\varepsilon_0 r} + \oint\!\!\oint_S \frac{\mathrm{d}q_{感}}{4\pi\varepsilon_0 R}$$

$$= \frac{q}{4\pi\varepsilon_0 r}$$



不带电导体球



不带电导体球

方法二：电像法

定解条件：

q 给定

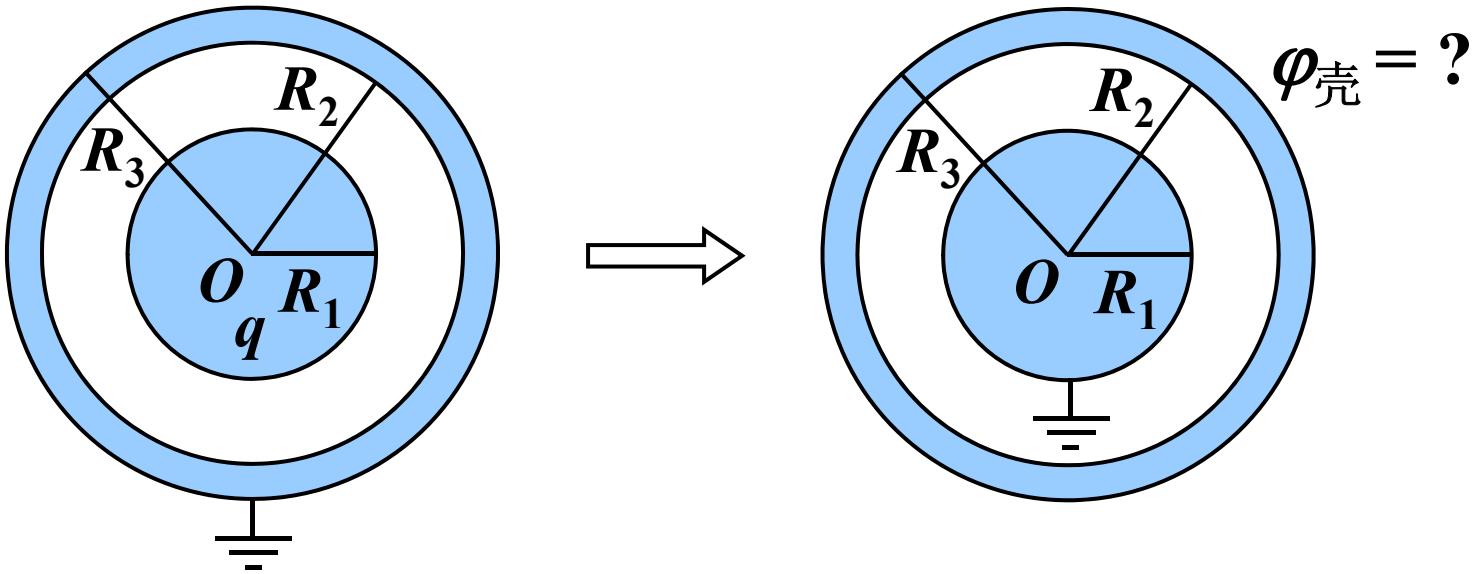
边界条件 $\begin{cases} \varphi_\infty = 0 \\ \text{导体 } Q = 0 \end{cases}$

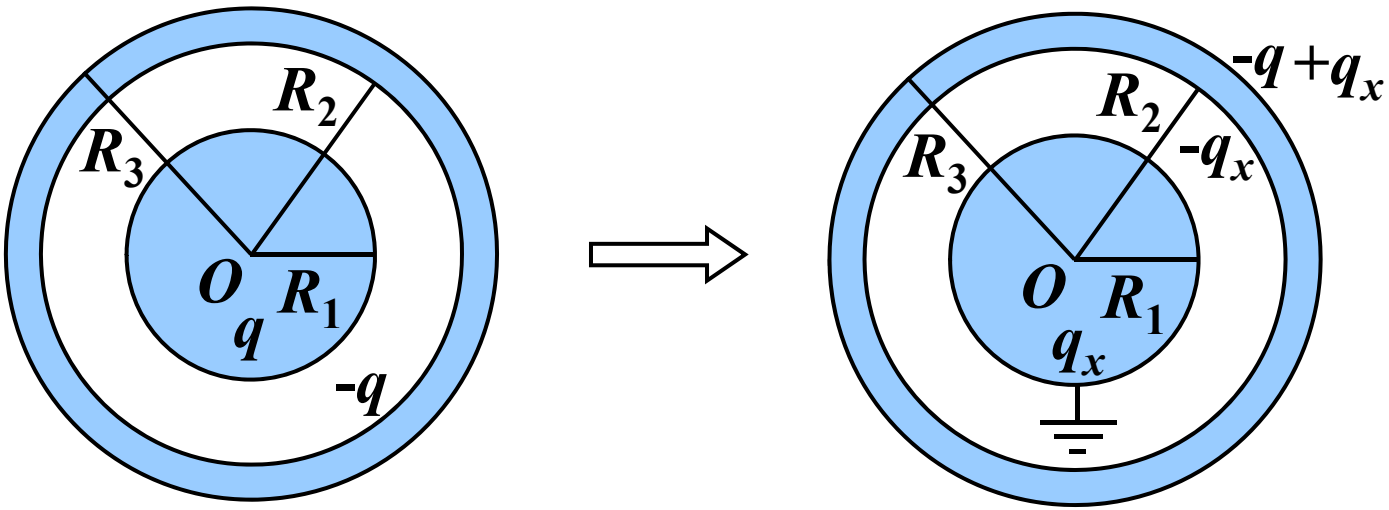
电像放置 $\left\{ \begin{array}{l} r' = \dfrac{R^2}{r},\quad q_1' = -\dfrac{R}{r}q \\ \text{球心 } O \text{ 处},\quad q_2' = -q_1' \end{array} \right\}$ 满足定解条件

$$\varphi = \frac{q_2'}{4\pi\varepsilon_0 R} = \frac{1}{4\pi\varepsilon_0 R}\cdot\frac{R}{r}q = \frac{q}{4\pi\varepsilon_0 r}$$

7. 电量 q、半径 R_1 的导体球外部同心地套内、外半径为 R_2、R_3 的不带电导体球壳。把导体球壳接地后再与地绝缘，然后把导体球接地。

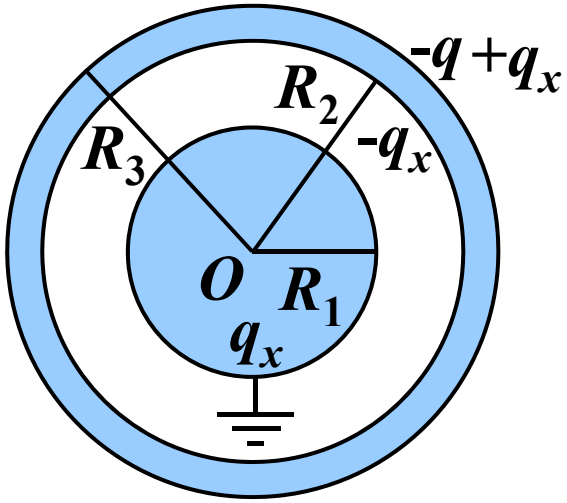
求：导体球壳的电势 $\varphi_{壳}$。





【思考】导体球接地后，其上有无电荷？

球接地前，壳内表面带电 $-q$。球接地后若不带电，壳内表面电量应变为零，$-q$ 电荷应移至壳外表面。则壳内空间电场为零，壳的电势与球相等，为零。这和壳外表面带电 $-q$， 相比无穷远电势为负矛盾。所以球接地后带电，设带电量为 q_x。



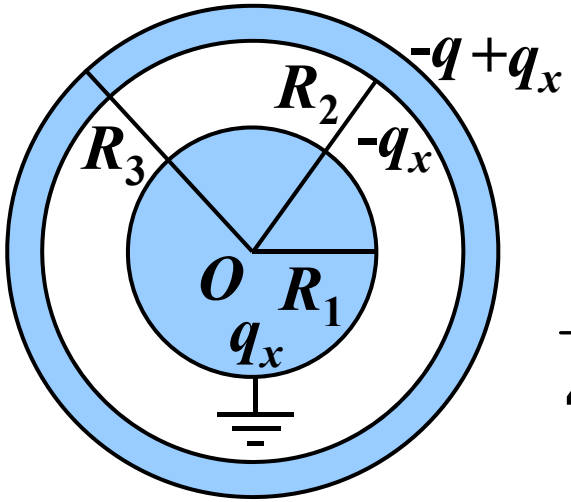
解： 方法一：场强积分法

$$\varphi_{壳}=\int_{R_2}^{R_1}\frac{q_x\,\mathrm{d}r}{4\pi\varepsilon_0 r^2}=\int_{R_3}^{\infty}\frac{(q_x-q)\,\mathrm{d}r}{4\pi\varepsilon_0 r^2}$$

$$\Rightarrow\ q_x=\frac{R_1R_2\cdot q}{R_1R_2+R_2R_3-R_1R_3}$$

$$\varphi_{壳}=\int_{R_2}^{R_1}\frac{q_x\,\mathrm{d}r}{4\pi\varepsilon_0 r^2}=\frac{q_x}{4\pi\varepsilon_0}\left(\frac{1}{R_2}-\frac{1}{R_1}\right)$$

$$=\frac{-(R_2-R_1)\cdot q}{4\pi\varepsilon_0(R_1R_2+R_2R_3-R_1R_3)}$$



方法二：电势叠加法

导体球电势：

$$\frac{-q+q_x}{4\pi\varepsilon_0 R_3}+\frac{-q_x}{4\pi\varepsilon_0 R_2}+\frac{q_x}{4\pi\varepsilon_0 R_1}=0$$

$$\Rightarrow\ q_x=\frac{R_1R_2\cdot q}{R_1R_2+R_2R_3-R_1R_3}$$

$$\varphi_{壳}=\frac{-q+q_x}{4\pi\varepsilon_0 R_3}=\frac{-(R_2-R_1)\cdot q}{4\pi\varepsilon_0(R_1R_2+R_2R_3-R_1R_3)}$$